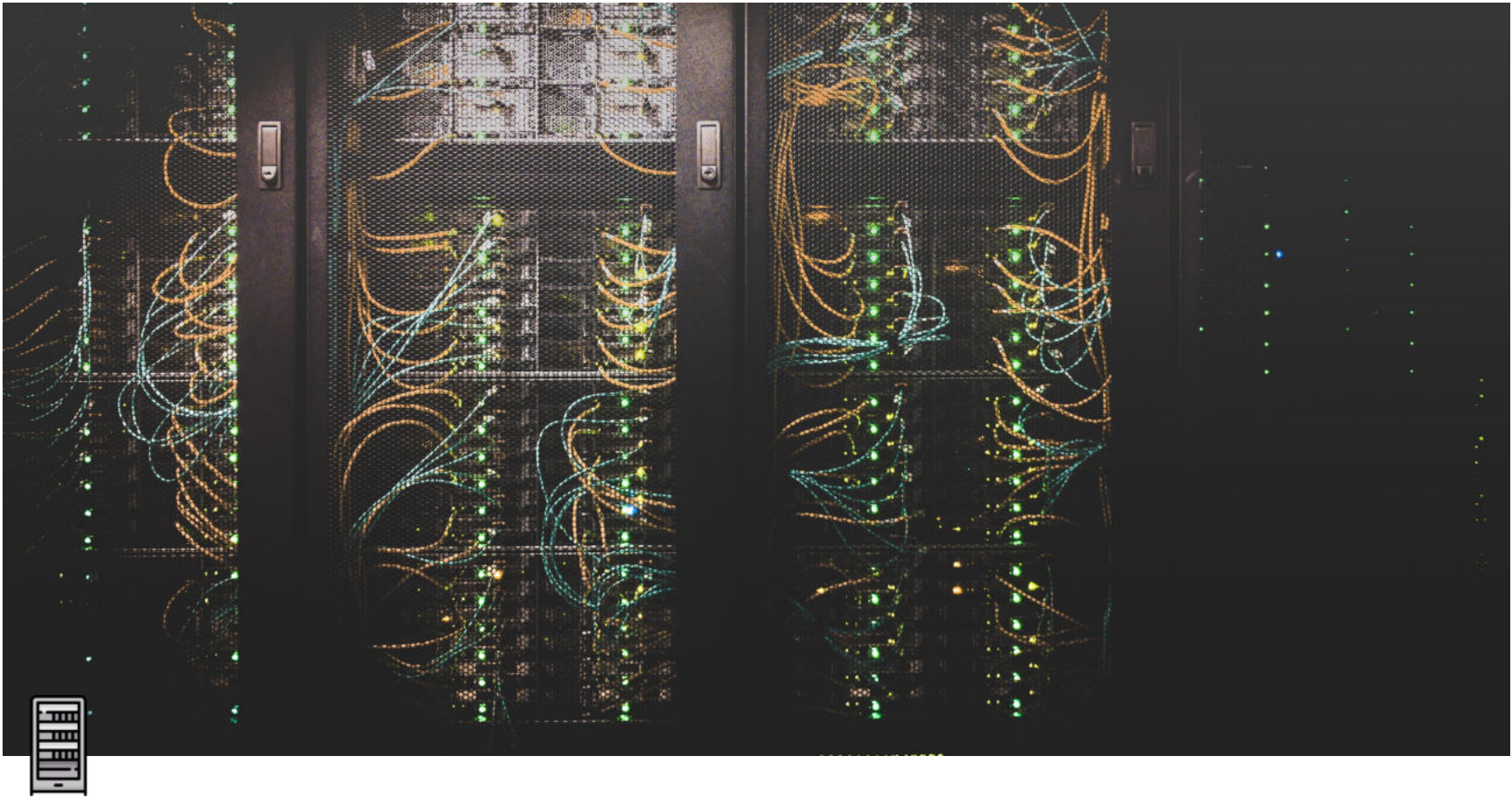




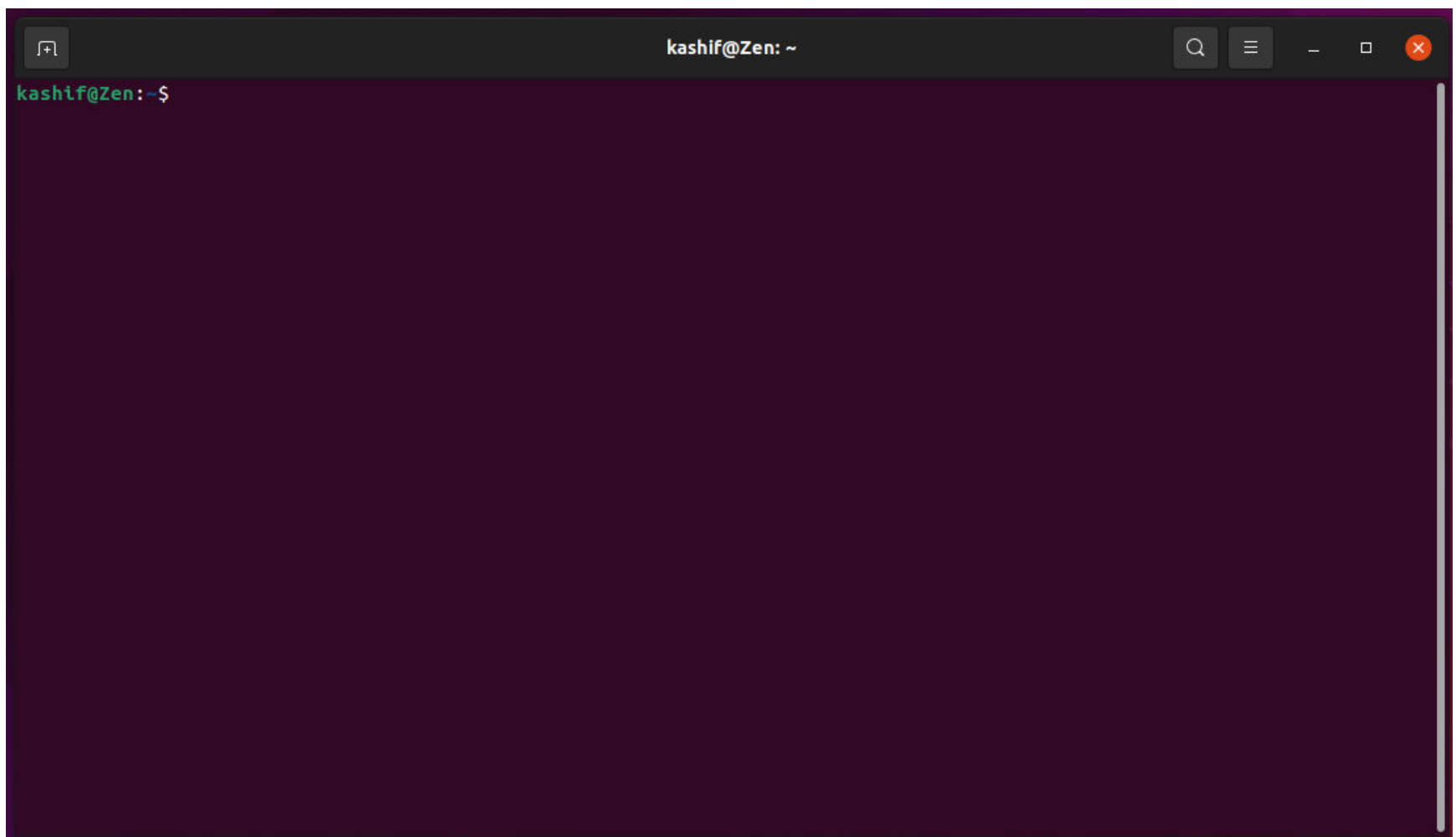
Command Line Environment

Type	Lecture
Date	@December 22, 2021
Lecture #	2
Lecture URL	https://youtu.be/qrAnIpcMyYc
Notion URL	https://21f1003586.notion.site/Command-Line-Environment-f96a91e782a147a88894028d7848a2c4
Week #	1

Why use Command Line environment?

- To use linux at its max potential
- Combine commands to form powerful scripts
 - To automate using these scripts
- *To assert dominance over GUI plebs*

Terminal in Ubuntu



This is what the default terminal looks like in Ubuntu

To clear the command line

```
clear
```

- or you can press `Ctrl + L` to clear the terminal screen

To check which directory we currently are in

```
pwd
```

By default, you are placed in the home directory of the currently logged in user

To list all the files and folders in the current directory

```
ls
```

To view the currently running processes

```
ps
```

To know the OS, duh

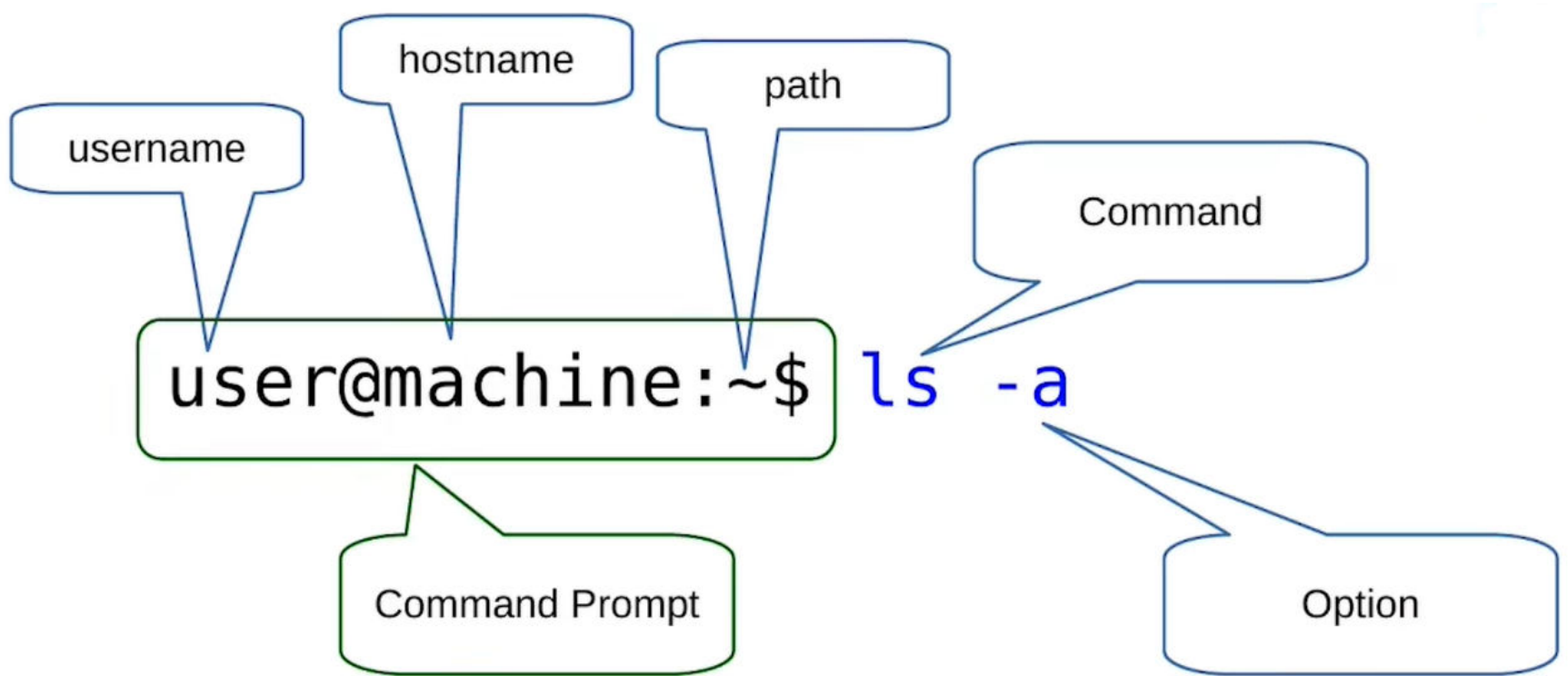
```
uname
```

To exit the shell

```
exit
```

- Or you can also press `Ctrl + D` to exit out of the terminal session

Anatomy of a typical command on the terminal



- To display all the files, press `ls -a`

A screenshot of a terminal window titled "kashif@Zen: ~". The command `kashif@Zen:~$ ls -a` has been executed. The output lists files and directories in two rows: `.`, `.bash_history`, `.bashrc`, `.config`, `Documents`, `.gitconfig`, `Music`, `.profile`, `snap`, `Templates` in the first row, and `..`, `.bash_logout`, `.cache`, `Desktop`, `Downloads`, `.local`, `Pictures`, `Public`, `.sudo_as_admin_successful`, `Videos` in the second row. The prompt `kashif@Zen:~$` is followed by a cursor.

- To display the files in a list, press `ls -l`

```
kashif@Zen: ~  
kashif@Zen:~$ ls -l  
total 36  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Desktop  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Documents  
drwxr-xr-x 3 kashif kashif 4096 Dec 21 20:16 Downloads  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Music  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Pictures  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Public  
drwx----- 3 kashif kashif 4096 Dec 21 20:15 snap  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Templates  
drwxr-xr-x 2 kashif kashif 4096 Dec 21 19:58 Videos  
kashif@Zen:~$ |
```

These two flags are the most commonly used ones, **we can also combine them as one flag** like, `ls -al`

To get help on any command

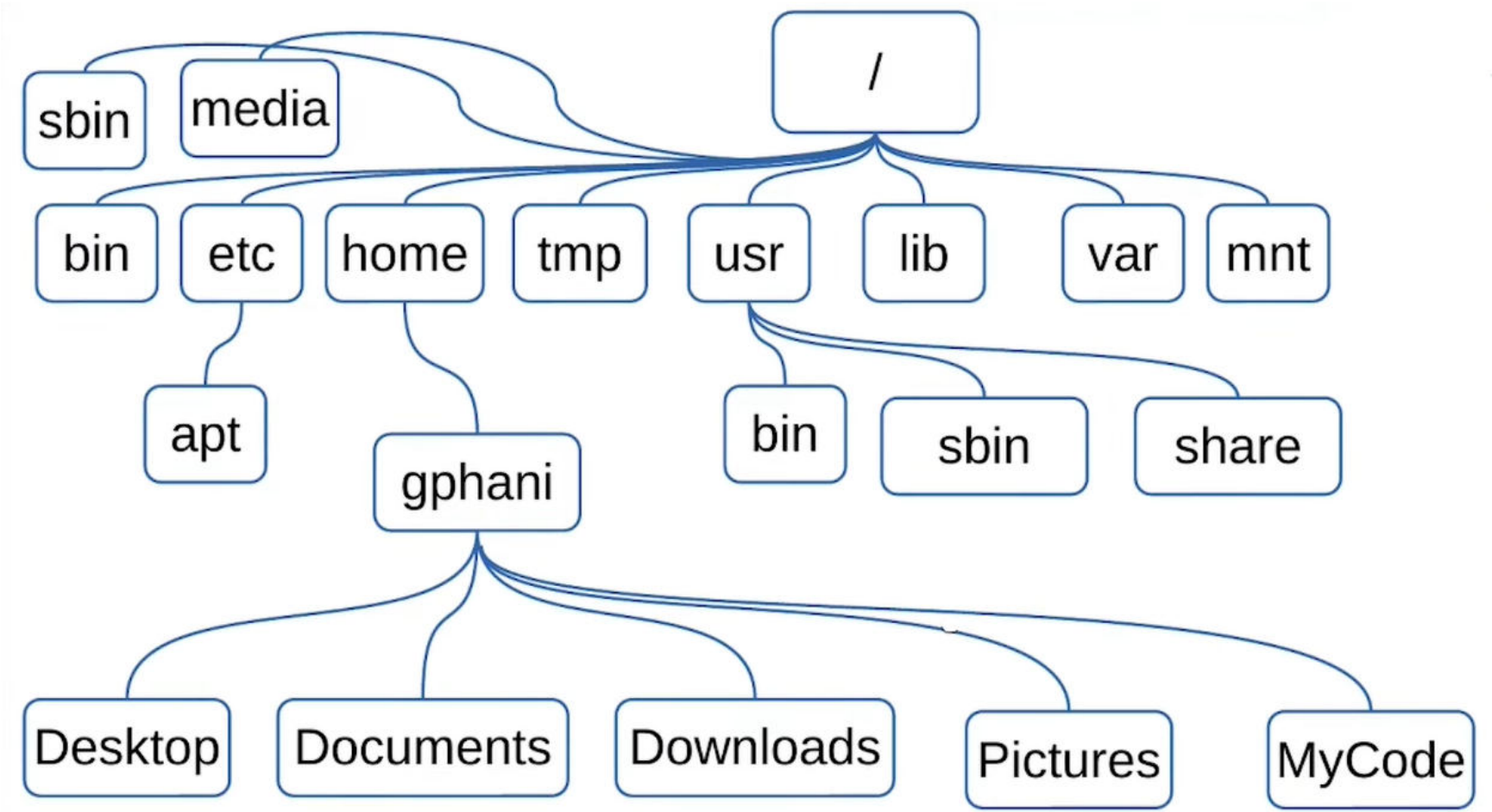
```
man <command-name>
```

- For example

```
man ls
```

gives us the manual for the command `ls`

Filesystem in Linux



Traversing the tree

`/` is the root of the file system

`/` is also the delimiter for sub-directories

`.` is the current directory

`..` is the parent directory

Path for traversal can be absolute or relative

To change directory

```
cd <location>
```

Examples

- `cd` without any arguments will take us to the home directory of the currently logged in user
- `cd <folder-name>` will change the move us into the `folder-name`
- `cd ..` will takes us to the parent of the current directory
- `cd /` will takes us to the root directory

What does this *directory* do?

- `/bin` → Essential command binaries
- `/boot` → Static files for the bootloader
- `/dev` → Device files
- `/etc` → Host specific system configuration
- `/lib` → Essential shared libraries and kernel modules
- `/media` → Mount points for removable devices
- `/mnt` → Mount points
- `/opt` → Add on application software packages
- `/run` → Data relevant to running processes
- `/sbin` → Essential system binaries
- `/srv` → Data for services
- `/tmp` → Temporary files
- `/usr` → Secondary hierarchy
- `/var` → Variable data

`/usr` hierarchy

- `/usr/bin` → User commands
- `/usr/lib` → Libraries
- `/usr/local` → Local hierarchy
- `/usr/sbin` → Non-vital system binaries
- `/usr/share` → Architecture dependent data
- `/usr/include` → Header files included by C programs
- `/usr/src` → Source code

`/var` hierarchy

- `/var/cache` → Application cache data
- `/var/lib` → Variable state information
- `/var/local` → Variable data for `/usr/local`
- `/var/lock` → Lock files

- `/var/log` → Log files and directories
- `/var/run` → Data relevant to running processes
- `/var/tmp` → Temporary files preserved between reboots

	sharable	unsharable
static	/usr /opt	/etc /boot
variable	/var/mail	/var/run /var/lock